

Snatcher: Apple Find My Network Exposes Your Lost Devices to Strangers

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A Artifact Appendix

A.1 Abstract

Snatcher demonstrates that the mechanisms making Apple’s Find My network effective for owners also expose lost Apple devices to unauthorized discovery, tracking, and physical theft by strangers. The attack relies solely on a commodity Android smartphone and requires no specialized hardware. The paper formalizes a progressive three-level attack: (*Level 1*) Acoustic direction finding; (*Level 2*) Passive RSSI-IMU navigation; and (*Level 3*) De-anonymization via spatial-temporal clustering.

The released artifact includes the SnatchAPP Android application for discovering and actuating nearby lost devices, two ESP32 firmware images as evaluation targets, and representative raw data traces with offline analysis code to verify the navigation attacks safely. The artifact is packaged as a Git repository (also archived on Zenodo), is licensed under AGPL-3.0, and targets the *Artifacts Available*, *Artifacts Evaluated (Functional and Reusable)*, and *Results Reproduced* badges.

A.2 Description & Requirements

The repository is laid out as follows:

Snatcher/	
SnatchAPP/	# sound-play app (paper Table 1)
esp32_airtag_mac_rotation/	# mimicked AirTag (paper Fig 11)
esp32_sound_maker/	# sound trigger fw (paper Table 1)
reproduce/	# example traces + analysis (Python)
1_timeout_disconnect_distance/	# lost-state (paper Fig 3)
2_mac_rotation/	# MAC rotation (paper Table 2)
3_acoustic_direction/	# Level 1 direction (paper Fig 6)
4_rssi_navigation/	# Level 2 RSSI-IMU nav (paper Fig 8)
5_clustering/	# Level 3 clustering (paper Fig 11)
6_max_distance/	# attack range (paper Fig 14,15)
7_navigation_traces/	# end-to-end traces (paper Fig 17)
doc/	# this artifact appendix
README.md LICENSE	# quick start + AGPL-3.0

The top-level README gives a quick start. SnatchAPP and esp32_sound_maker actuate the non-owner sound trigger (paper Table 1); esp32_airtag_mac_rotation is the mimicked-AirTag target for Level-3 clustering. The reproduce/ folder holds seven self-contained examples and is the primary, hardware-free verification path: each pairs a small data/ trace with Python

scripts that print a result and write a figure, reproducing one named result of the paper (named in the tree above and in each experiment below). In total they reproduce the paper’s **Figures 3, 6, 8, 11, 14, 15, and 17** and **Tables 1 and 2** — the lost-state threat model, the three attack levels, the maximum attack range, and the end-to-end navigation cost.

A.2.1 How to access.

- **Archived (permanent):** Zenodo concept DOI (resolves to the latest version), <https://doi.org/10.5281/zenodo.20693210>.
- **Development version:** <https://github.com/rzy0901/Snatcher>.
- **Anonymized mirror:** <https://anonymous.4open.science/r/Snatcher-4D25>.

A.2.2 Hardware dependencies.

- **Trace-based reproduction (E1–E7):** none — any commodity PC with Python suffices.
- **Attacker device (E0):** one Android smartphone running Android 7.0 (API 24) or newer, with Bluetooth Low Energy, a stereo microphone (the UNPROCESSED audio source must be supported for Level 1), and a 9-axis IMU.
- **Evaluation targets (E0):** two ESP32-family boards (we used the ESP32-WROVER-Kit; ESP32-S3/C3 also work) — one for the mimicked-AirTag firmware, one for the sound trigger. Optionally an AirTag/AirPods (Level 1) and an iPhone/Apple Watch (Level 2), *owned by the evaluator, signed into an Apple ID and placed in the Separated (Lost) state*, to validate against production hardware.
- USB cables to flash the ESP32 boards and to retrieve logs from the phone. No GPU is required.

A.2.3 Software dependencies.

- **Python 3.8+** with numpy, scipy, pandas, and matplotlib (reproduce/requirements.txt).
- *Optional, to build the live components:* Android Studio¹ (Hedgehog 2023.1.1+, compileSdk 34, JDK 17, adb) and ESP-IDF² v4.4+.

¹<https://developer.android.com/studio>

²<https://github.com/espressif/esp-idf>

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- The Android build pulls MPAndroidChart and AndroidX via Gradle; the ESP32 firmware derives from OpenHaystack (mimicked AirTag) and AirGuard (sound trigger).

A.2.4 Benchmarks. All traces are self-collected with SnatchAPP, which logs synchronized CSV streams—BLE scans (MAC, RSSI, payload, timestamp), IMU samples (accelerometer/gyroscope/magnetometer, fused orientation, PDR position, step count), and navigation RSSI (RSSI, phone heading, reconstructed position, target heading)—plus raw stereo PCM audio with per-recording metadata. Each example ships, in its data/ folder, the processed trace it consumes, so the analysis runs without collecting new measurements. The bulkier raw scan logs behind these traces are retained locally (available on request), while 1_timeout_disconnect_distance/, 2_mac_rotation/, and 6_max_distance/ also ship a representative raw capture under raw/ so the raw-to-trace derivation can be inspected. No personal or third-party user data are included.

A.2.5 Security, privacy, and ethical concerns. Snatcher is offensive — it can discover, actuate, and physically localize Apple Find My devices that are not the operator’s. Evaluate it *only* against devices you own (commercial Apple devices you control, or the provided ESP32 lab targets); never trigger sound on, connect to, or track a third party’s device. To avoid misuse, the released SnatchAPP omits the on-device navigation- and clustering-assistance modules; the trace-based reproduction still validates those algorithms offline from the shipped logs and performs no live attack. Running it is otherwise safe: no destructive steps, no privileged access, no security mechanisms disabled.

A.3 Set Up

A *prebuilt* SnatchAPP APK (part (A)) lets reviewers run the live on-device attack (E0) in about a minute with no build — *we encourage trying it*. Building from source (parts (A)/(B)) is optional, and the trace-based reproduction (E1–E7) runs from the provided traces and Python code (part (C)) on any PC.

A.3.1 Installation. (A) Android application. The fastest path is the *prebuilt, signed APK* — *no build needed*: download SnatchAPP.apk³ and install it on any Android 7.0+ phone (open it on the device and allow “install unknown apps”, or adb install -r SnatchAPP.apk). To build from source instead, open SnatchAPP/ in Android Studio (*Run* installs it), or:

```
cd SnatchAPP
./gradlew assembleRelease
adb install -r app/build/outputs/apk/release/app-release.apk
```

On first launch, grant the requested permissions (Bluetooth scan/connect, fine location, microphone, physical activity, and storage). Figure 1 shows the partially-released app running from Android Studio.

(B) ESP32 firmware (optional). For each firmware, source the ESP-IDF environment and, via idf.py menuconfig, set the Bluetooth controller to *BLE Only* and enable Bluedroid. Then build, flash, and monitor:

```
cd esp32_airtag_mac_rotation # or esp32_sound_maker
idf.py set-target esp32
```

³<https://github.com/rzy0901/Snatcher/raw/main/SnatchAPP/SnatchAPP.apk>

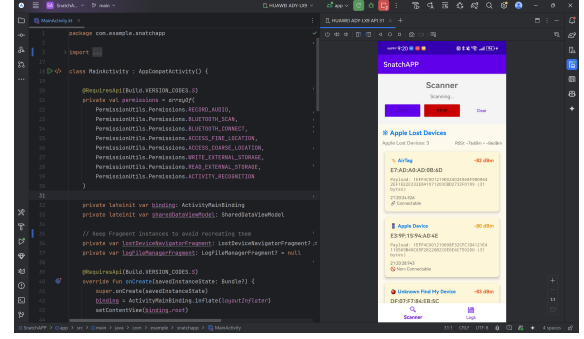


Figure 1: The partially-released SnatchAPP built and run from Android Studio. The Scanner screen lists nearby Apple lost devices with their type, MAC, RSSI, and Find My payload. Only lost-device discovery and the non-owner sound trigger are included; the navigation- and clustering-assistance modules are withheld to mitigate misuse.

```
--- (a) Mimicked AirTag: DEVICE_ID + 30 s MAC rotation ---
open_haystack: Device ID (ground truth): 0x01
open_haystack: MAC shuffle period: 30 seconds
open_haystack: using device address: fc ff ff ff 13 00
open_haystack: MAC shuffle timer started (period: 30 s)
open_haystack: Generated random MAC: dc 73 62 6a 46 db
open_haystack: MAC address shuffled, adv. restarts
open_haystack: Generated random MAC: dc 20 3c ac 11 bf
open_haystack: MAC address shuffled, adv. restarts
--- (b) Sound trigger: non-owner sound on a real AirTag ---
Connected, start service discovery...
AirTag service found: start=0x005D end=0xFFFF
Characteristic found, handle=0x005F. Sending sound play
AirTag: Sent sound play command (value=0xAF)
AirTag: Sound playback command sent successfully
```

Figure 2: ESP32 firmware serial logs. (a) the mimicked-AirTag firmware advertises under a ground-truth DEVICE_ID and rotates its MAC every 30 s; (b) the sound-maker firmware connects to a real AirTag and sends the non-owner sound-play command (paper Table 1). Full screenshots: doc/figs/fw_mac_rotation.png, fw_soundplay.png.

```
idf.py menuconfig # set BLE Only + Bluedroid
idf.py -p <PORT> flash monitor
```

Before building, each firmware takes a one-line edit:

- *Mimicked AirTag* (openhaystack_main.c): set DEVICE_ID (labels co-located devices) and the MAC-rotation period (paper: 10–60 s).
- *Sound trigger* (main.c): set device_type and target_mac to the victim accessory (read its MAC with SnatchAPP or nRF Connect⁴).

The serial monitor confirms each firmware runs (Figure 2): the mimicked AirTag shuffles its MAC on schedule, and the sound trigger plays a non-owner sound on a real Apple accessory (AirTag or AirPods; paper Table 1).

(C) Reproduction environment.

```
cd reproduce
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
```

⁴<https://www.nordicsemi.com/Products/Development-tools/nRF-Connect-for-mobile>

A.3.2 Basic test. After installing the reproduction environment, run the acoustic example as a quick functionality check:

```
cd reproduce/3_acoustic_direction && python acoustic_example.py
```

It prints the detected arrival level (≈ -42.8 dB), confirming the toolchain is installed. With the prebuilt APK, E0 below is the live demo.

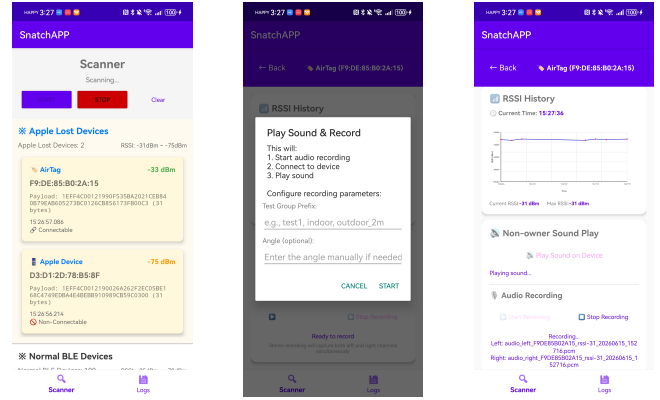
A.4 Evaluation Workflow

The evaluation is one functional check (E0) and seven reproduction experiments (E1–E7), which run from the provided data with no Apple hardware; the recommended E0 demo needs only an Android phone with the prebuilt APK. The aggregate field results (per-level success rates, paper Figure 16, and averaged end-to-end navigation times) need extensive real-world trials and are out of automated scope; the artifact provides the implementation and logging to repeat them.

A.4.1 Major claims.

- (C1): A separated device becomes discoverable quickly and at area scale: it enters its advertising lost state within minutes of the owner leaving, and the owner’s BLE link drops at ~ 10 – 20 m across diverse indoor scenes (paper Figure 3). Validated by (E1).
- (C2): Find My MAC-address rotation is slow enough to track: Apple *accessories* keep one address for ~ 24 h, versus a fixed ~ 15 min (iPhone) or ~ 19 – 36 min (Apple Watch) – Vulnerability III, paper Table 2. Validated by (E2).
- (C3): A non-owner can find the *direction* to a lost accessory that supports non-owner sound play (e.g., AirTag, AirPods) by triggering its sound and comparing per-orientation loudness (Level 1, paper Figure 6). Validated by (E3).
- (C4): A *silent* lost device can be navigated to by fusing its BLE RSSI with phone IMU/PDR (Level 2, paper Figure 8). Validated by (E4).
- (C5): The rotating MAC addresses of co-located lost devices can be re-associated into per-device clusters, defeating MAC-address randomization (Level 3, paper Figure 11). Validated by (E5).
- (C6): The attack operates at practical range: a lost device stays BLE-discoverable out to tens of metres, and the non-owner sound trigger is effective at close range, both degrading gracefully with distance (paper Figures 14–15). Validated by (E6).
- (C7): Under all three levels the attacker’s reconstructed walking trajectory converges on the target, as sampled from the end-to-end navigation evaluation (paper Figure 17). Validated by (E7).

A.4.2 Experiments. (E0) Live on-device demo of the attack primitives [*strongly recommended*; ~ 10 person-min; just install the APK]. Install the prebuilt SnatchAPP APK (part (A)), open the *Scanner*, and watch it discover the *real* Apple devices in the Separated (Lost) state broadcasting around you (usually several in any public space). Then, against a target you control (an AirTag/AirPods you own placed in Lost mode, or the provided ESP32 mimicked-AirTag), confirm each primitive operates end to end:



(a) *Scanner*: tap a connectable lost device.

(b) *Play Sound & Record*: record, connect, play.

(c) The alert plays; stereo audio is recorded.

Figure 3: E0, non-owner sound trigger (Vuln. II). On the *Scanner* a connectable lost device is tapped (a); *Play Sound & Record* (b) starts a stereo recording, connects with no authentication, and plays the alert (c), saving `audio_left/right_*.pcm`.

- (1) **Cleartext discovery (Vuln. I).** On the *Scanner*, verify each lost device is detected and identified from its Find My advertisement (the app filters the FF 4C 00 12 19 prefix and decodes the device-type byte, e.g. AirTag, AirPods).
- (2) **Unauthenticated actuation (Vuln. II).** Tap a connectable lost device, open its detail view, and choose *Play Sound & Record*: with no pairing or authentication, the app records stereo audio, connects, and plays the non-owner alert sound (Figure 3; confirm it sounds on a real AirTag/AirPods or the sound-trigger ESP32). The recording (`audio_left/right_*.pcm` + a `*_metadata` file) is the Level-1 acoustic input for E3 (top mic = right channel).
- (3) **Passive RSSI/IMU logging (Vuln. III).** On the data-collection screen, confirm the app records the lost device’s RSSI together with the phone’s IMU motion, producing the synchronized logs that E4/E5 consume.

Expected: the device is discovered, the alert plays, and the session writes its outputs under the app’s external files directory – the BLE/IMU logs (`ble_scan_*.csv`, `imu_data_*.csv`, `lost_device_*.csv`, `NavRSSI_*.csv`) under Documents/SnatchAPP_Logs, and the sound recordings (`audio_left/right_*.pcm`) under Music – demonstrating the artifact is *Functional*. Copy them off most simply by connecting the phone over USB and opening either folder in a file browser (Figure 4), or via ADB:

```
adb pull /sdcard/Android/data/com.example.snatchapp/files/Documents/
  SnatchAPP_Logs
adb pull /sdcard/Android/data/com.example.snatchapp/files/Music
```

(E1) Lost-state onset (paper Figure 3) [5 person-min]. How quickly, and from how far, a separated device starts advertising in its lost state – the precondition for discovery. **Preparation:** `cd reproduce/1_timeout_disconnect_distance`.

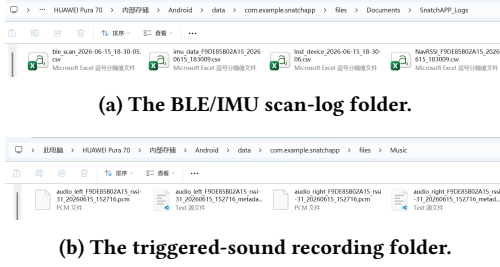


Figure 4: E0 output retrieval without ADB: with the phone connected over USB, both the scan logs (a) and the triggered-sound recordings (b) are visible directly in a file browser and can be copied off.

Execution: `python plot.py`.

Results: a two-panel figure (Figure 5). (a) the lost-state timeout per device on a log axis — AirTag ~12 min, AirPods longer but variable, Apple Watch ~13 s; (b) the owner-to-device disconnect distance stays at ~10–20 m across four indoor scenes. These reproduce Figure 3 of the paper.

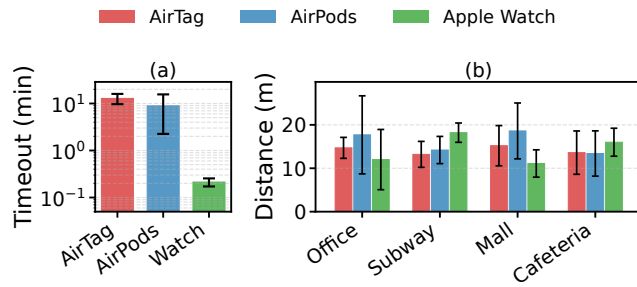


Figure 5: E1, lost-state onset (paper Figure 3): per-device time to enter the lost state (a, log axis) and the owner-to-device disconnect distance across four indoor scenes (b).

(E2) MAC-address rotation period (paper Table 2) [5 person-min]. The Find My address-rotation period per device — Vulnerability III, which bounds how long one address can be followed. **Preparation:** `cd reproduce/2_mac_rotation`.

Execution: `python make_table.py`.

Results: a per-device table (Table 1) that reproduces paper Table 2: Apple accessories hold one MAC for ~24 h, while iPhone and Apple Watch rotate every ~15–36 min. `make_table.py` also writes `mac_rotation.pdf` (a log-scale boxplot of the per-address spread), and `derive_durations.py` re-derives the iPhone column from the shipped capture. The long static window on accessories is what makes Level-3 tracking practical.

Table 1: E2 output, reproducing paper Table 2: measured Find My MAC-address rotation period per device.

Category	Device	MAC rotation period
Apple Accessories	AirPods Pro	~24 h (daily, ~4 am)
	AirTag	~24 h (daily, ~4 am)
Apple Devices	iPhone	fixed ~15 min
	Apple Watch	19–36 min

(E3) Acoustic direction finding (Level 1, paper Figure 6) [10 person-min + <5 compute-min]. The input is the phone’s top-mic channel (`data/example_top.pcm`, 16-bit mono PCM at 44.1kHz) and the model chirp template (`data/template.wav`).

Preparation: `cd reproduce/3_acoustic_direction`.

Execution:

```
python plot_waveform_spectrogram.py # waveform + spectrogram
python acoustic_example.py          # arrival detection + dB
python plot_direction_polar.py      # direction polar
```

Results: `plot_waveform_spectrogram.py` shows the chirp energy around 2.6–2.8 kHz (Figure 6a). `acoustic_example.py` band-passes both signals to that band, cross-correlates to detect the arrivals, and takes the RMS over each in dB (≈ -42.8 dB; Figure 6b). The four body orientations give four dB values, which `plot_direction_polar.py` turns into the contrast-enhanced score $S_k = \alpha A_k + \beta (A_k - A_{opp(k)})$ ($\alpha=0.3, \beta=0.7$); the heading $\arg \max_k S_k$ falls at the *Front* by body shadowing (Figure 6c), reproducing the paper’s Level 1 result.

(E4) RSSI-IMU navigation (Level 2, paper Figure 8) [10 person-min]. SnatchAPP logs, per BLE reading, the target’s RSSI with the phone heading and the IMU-derived Pedestrian-Dead-Reckoning (PDR) position and step count (column format in the folder’s README); two sample logs are provided. **Preparation:** `cd reproduce/4_rssi_navigation`.

Execution:

```
python plot_trace.py                # data/nav_trace_1.csv
python plot_trace.py --log data/nav_trace_2.csv
```

Results: the script picks the target device (the frequently-seen MAC with the highest peak RSSI), reconstructs the walked path from the logged PDR position, and colours it by RSSI. The path converges on the target as RSSI rises (e.g. $-81 \rightarrow -39$ dBm; Figure 7), reconstructing the paper’s Level 2 navigation trace.

(E5) Spatial-temporal clustering (Level 3, paper Figure 11) [10 person-min]. Co-located mimicked AirTags (the provided ESP32 firmware) at different distances. For exact ground truth the script applies the MAC-address rotation in software, each device rotating independently (the firmware does the same on-device); the clustering runs blind, on timing and RSSI only. **Preparation:** `cd reproduce/5_clustering`.

Execution:

```
python cluster.py --log data/clustering_log_3dev.csv # 3 devices
python cluster.py                                # 2 devices
```

Results: the script runs the paper’s two stages. *Static profiling* colours the temporal-conflict graph — a device cannot advertise two MACs at once, so overlapping MACs seed different clusters — and reads off each device’s ~30 s rotation period. *Dynamic stitching* then attaches each later MAC to the one cluster whose previous MAC has already ended, breaking ties by RSSI and period. On the three-device log (Figure 8), whose RSSI ranges overlap and cross, the temporal constraint still recovers all three trajectories, defeating MAC rotation as in the paper’s Level 3 result.

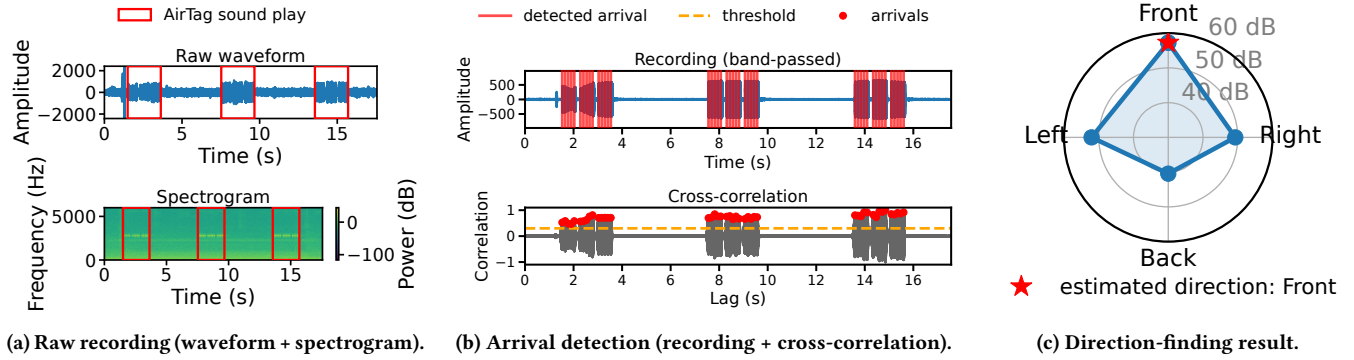


Figure 6: E3, acoustic direction finding (Level 1): the top-mic recording (a), band-passed and cross-correlated with the chirp template, gives the arrival level (≈ -42.8 dB) (b); four body orientations yield the direction estimate, here *Front* (c).

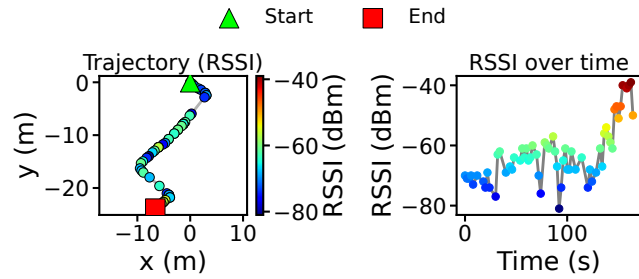


Figure 7: E4, navigation trace: the walked PDR trajectory coloured by RSSI (left) and RSSI over time (right), converging on the target as RSSI rises.

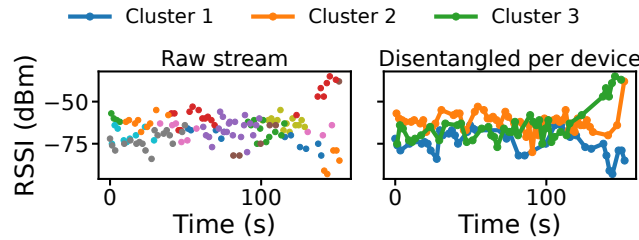


Figure 8: E5, clustering (three devices): from the fragmented rotated-MAC stream (left) the temporal-overlap + RSSI method recovers the three per-device trajectories (right).

(E6) Maximum attack distance (paper Figures 14–15) [10 person-min]. Two field sweeps over the attacker-to-device distance (0–80 m): the BLE discovery channel and the non-owner sound trigger. **Preparation:** `cd reproduce/6_max_distance`.

Execution: `python plot_max_distance.py`.

Results: two dual-axis plots (Figure 9): (a) RSSI decays while the scanned interval rises sharply beyond ~ 60 m (discoverable to tens of metres); (b) the triggered sound is loud near the device (~ 92 dB), fading to ambient by ~ 30 m — reproducing Figures 14–15.

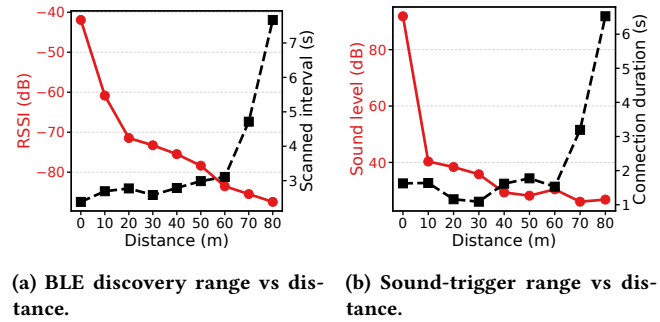


Figure 9: E6, maximum attack distance (paper Figures 14–15): the attack’s BLE-discovery range (a) and non-owner-sound range (b), swept from 0 to 80 m.

(E7) End-to-end navigation sample traces (paper Figure 17) [5 person-min]. Sample walking trajectories from the end-to-end evaluation, one per attack level. **Preparation:** `cd reproduce/7_navigation_traces`. **Execution:** `python plot_traces.py`.

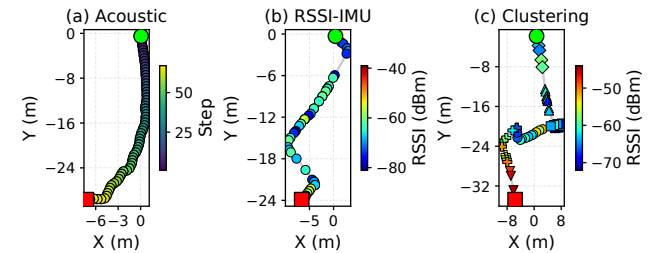


Figure 10: E7, sample end-to-end navigation traces (paper Figure 17), one per attack level, each homing from green Start to red End on the target.

Results: three sample trajectories (Figure 10), one per level: (a) Level-1 acoustic; (b) Level-2 RSSI-IMU (colour = RSSI); (c) Level-3 under clustering (MAC rotates every 30 s; colour = RSSI, shape = MAC). Each converges on the target (green Start $\rightarrow$ red End); panel (b) reuses the (E4) trace.